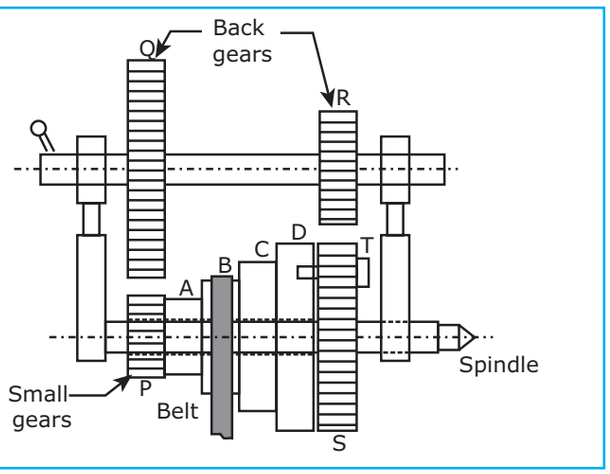


3D- Back gear mechanism



Back gear mechanism

Drive from step cone pulley

When the spindle gets from the cone pulley, the back gears Q and R are disengaged from the gears P and S. The pin 'T' is engaged with cone pulley. The belt can be arranged on the steps ABC or D to get four different direct speeds for the spindle.

YUEDPT

Drive through back gears

Back gears Q and R are engaged with gears P and S. The pin 'T' is disengaged from the cone pulley to make the cone pulley and the spindle separate units. When the cone pulley gets drive through the belt, the power is transmitted through the gears P, Q and R to the gear S, because of number of teeth on these gears, the spindle rotates at slower speeds. By arranging the belt on the different steps of the cone pulley, four different spindle speeds are obtained.

Uses of back gear arrangement

1. It is used for reducing the spindle speed, when turning on lager diameter of the work pieces and cutting coarse threads.
2. Eight different types of spindle speed can be obtained by this mechanism.

1.11 Gear Box Mechanism

Modern lathes are equipped with all geared headstocks to obtain different spindle speeds quickly. Casting of the all geared headstock has three shafts (1, 2 & 3) mounted within it. The intermediate shaft (2) has got three gears D, E and F and rotates at the same place and cannot move. The shaft (1) which is above the intermediate shaft has got three set of gears A, B and C mounted on it by keys. These three set of gears can be made to slide on the shaft with the help of a lever. This movement enables the gear A to have contact with the gear D or the gear B with gear E or the gear C with the gear F.

Likewise the spindle shaft (3) which is also splined has three set of gears G, H and I. These three set of gears can be made to slide on the shaft by lever. This sliding movement enables the gear G to have contact with gear D or the gear H with the gear E or the gear I with the gear F. By altering the position of the six gears namely A, B, C, G, H and I the following arrangements can be made within the headstock. Nice different spindle within the headstock. Nine different spindle speeds are obtained.

XII BME_English version CHAPTER 01.indd 11 1/8/2020 1:45:52 PM